CFD OPTIMIZATION REPORT

Project: CR-2026-B | Prepared for: Apex Architecture Group

1. EXECUTIVE SUMMARY

HyperFOAM performed CFD analysis for Apex Architecture Group to validate the HVAC design for a conference room with 12 occupants. The AI-driven inverse design optimizer determined optimal supply velocity (0.30 m/s) and diffuser angle (22.5 deg). Full 5-minute steady-state simulation confirms COMPLIANCE with ASHRAE Standard 55 thermal comfort criteria.

2. AI-OPTIMIZED SETTINGS

- Supply Air Velocity: 0.30 m/s (AI-optimized)
- Diffuser Angle: 22.5 deg (AI-optimized)
- Supply Temperature: 20.0 C
- Occupant Load: 12 persons x 100W = 1200W

3. PERFORMANCE METRICS

Metric	Value	Status
Temperature	21.58 C	COMPLIANT
CO2	406 ppm	COMPLIANT
Draft Velocity	0.217 m/s	COMPLIANT

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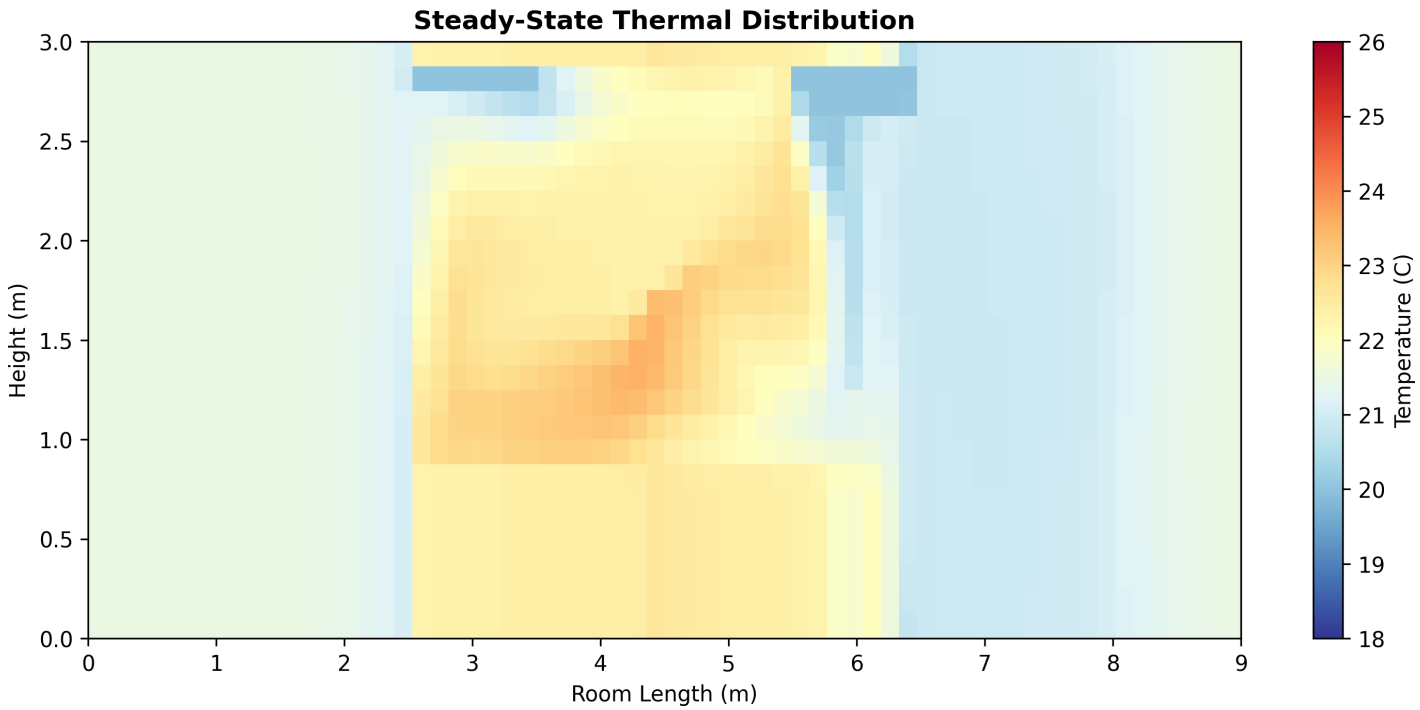
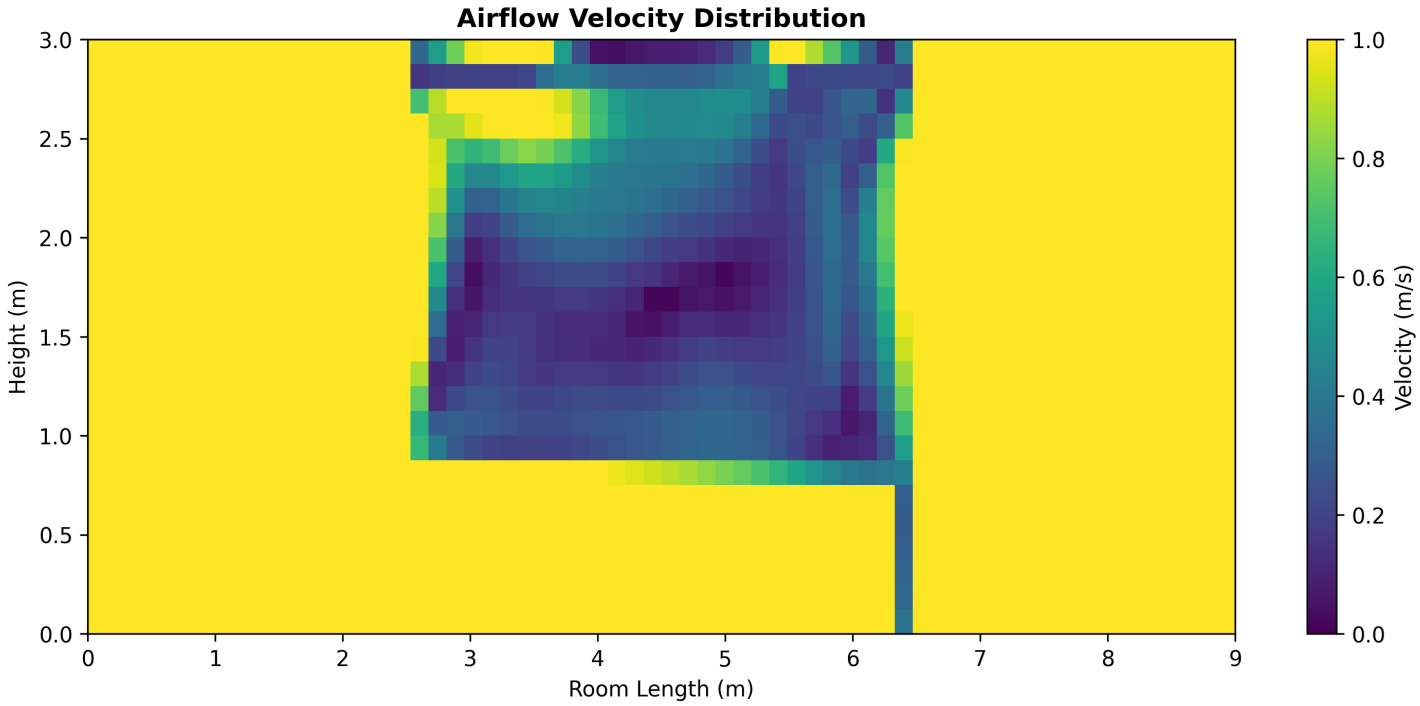
4. THERMAL ANALYSIS



5. AIRFLOW ANALYSIS



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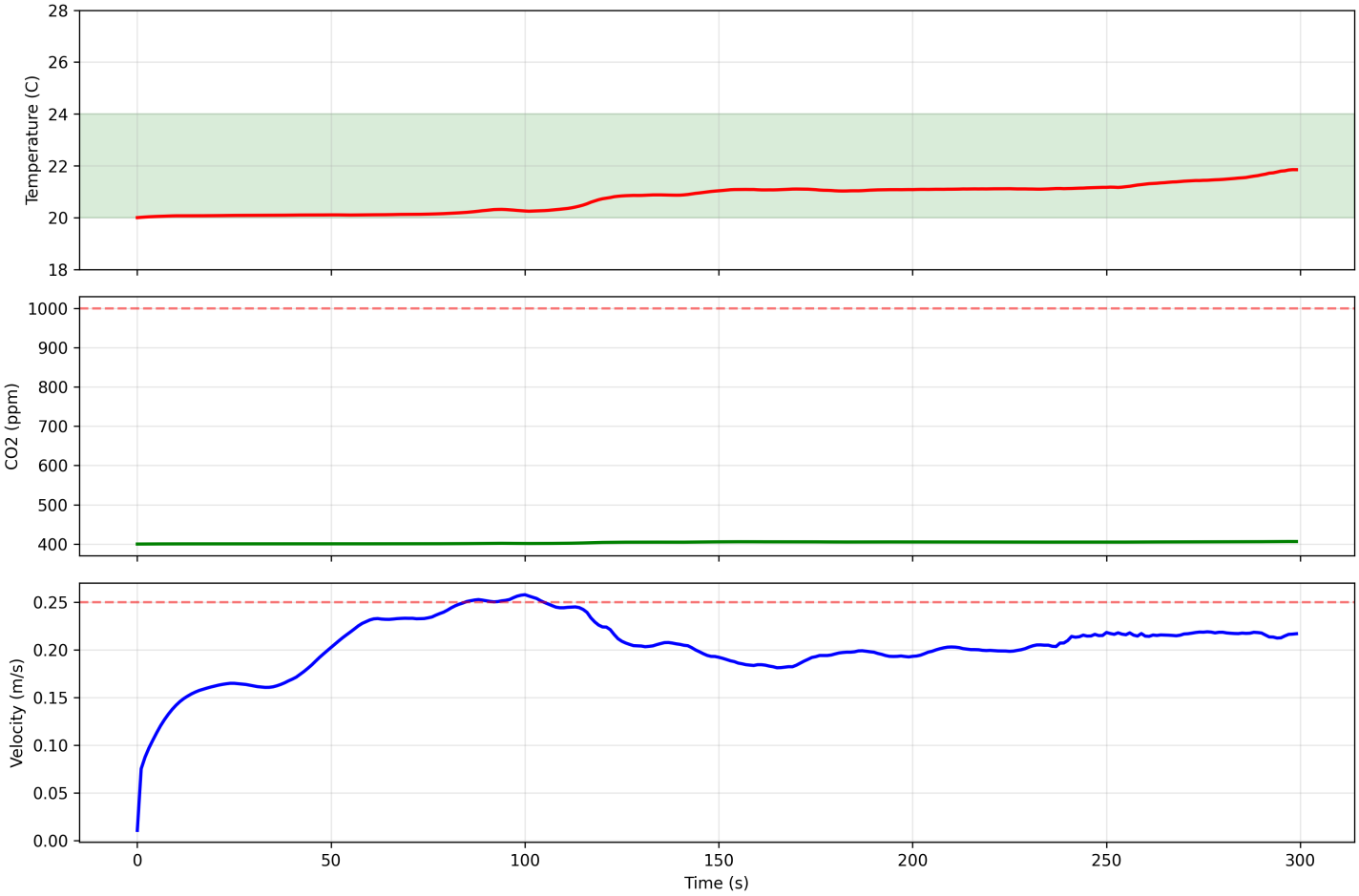
6. CONVERGENCE DATA



DISCLAIMER

This report is for preliminary engineering assessment only. Final equipment sizing and code compliance are the responsibility of the PE of Record.